

Teorija polj

Operatorji polj

s = skalarno polje, V = vektorsko polje

grad: $\nabla s = \left(\frac{\partial s}{\partial x}, \frac{\partial s}{\partial y}, \frac{\partial s}{\partial z}\right)$

div: $\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$

rot: $\nabla \times \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} =$

$= (Ry - Qz, Pz - Rx, Qx - Py)$

Laplacijan: $\Delta s = \nabla^2 s = \frac{\partial^2 s}{\partial x^2} + \frac{\partial^2 s}{\partial y^2} + \frac{\partial^2 s}{\partial z^2}$

Smerni odvod: $\frac{\partial s}{\partial l} = grad(s) \cdot \frac{\vec{l}}{|\vec{l}|}$

Lastnosti operatorjev polj:

$div\ grad\ f = \Delta f; \quad div\ rot\vec{v} = 0$

$di\ f(\vec{v}) = f\ div\ \vec{v} + grad\ f \cdot \vec{v}$

Prostorska geometrija

Koordinatni sistemi

Polarni: $x = r \cos \theta, \quad y = r \sin \theta \quad J = r$

Cilindrični:

$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z \quad J = r$

Sferični: $x = r \sin \phi \cos \theta, \quad y = r \sin \phi \sin \theta,$

$z = r \cos \phi \quad J = r^2 \sin \phi$

$J = det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}$

Razno formule

Ločna dolžina: $s = \int_a^b |\mathbf{r}'(t)|\, dt =$

$= \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\, dt$

Naravna parameterizacija:

$x'(s)^2 + y'(s)^2 + z'(s)^2 = 1$

paramet. daljice: $\vec{r}(t) = r_A * t(r_B - r_A)$

Odvod integrala s parametrom:

$F(x) = \int_{u(x)}^{v(x)} f(x, y) dy \Rightarrow$

$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial(f(x, y))}{\partial(x)} dy +$

$+ f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x)$

Tangentna premica in normalna ravnina:

$\vec{e}$ ali $\vec{n} = \dot{\vec{r}}(t_0)$

Kot med Vektorjema: $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$

Večkratni integrali

Dvojni integral:

$\iint_D f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$

Ploščina: $A = \iint_D 1\, dA$

Prostornina: $V = \iint_D f(x, y)\, dA$

Težišče: $\bar{x} = \frac{1}{A} \iint_D x\, dA, \quad \bar{y} = \frac{1}{A} \iint_D y\, dA$

Masa: $m = \iiint_E \rho(x, y, z)\, dV$

Vztrajnostni moment: $I = \iiint_E r^2 \rho\, dV$

Krivuljni integrali

S. polje: $\int_C f(x, y, z) ds =$

$= \int_{t_0}^{t_1} f(x(t), y(t), z(t)) * \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} dt$

v. polje: $\int_C \mathbf{V} \cdot d\mathbf{r} = \int_C (P dx + Q dy + R dz) =$

$= \int_{t_0}^{t_1} (P\dot{x} + Q\dot{y} + R\dot{z}) dt$

$d\vec{r} = \dot{\vec{r}} dt = (\dot{x} dt, \dot{y} dt, \dot{z} dt)$

Neodvisen od poti če $\vec{v}$ potencialno polje
($rot\vec{V} = 0$)

Ploskovni integrali

S. polje: $\iint_S f\, dS =$

$= \iint_D f(\mathbf{r}(u, v)) \sqrt{EG - F^2}\, du\, dv$

$E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$

$dS = \sqrt{1 + z_x^2 + z_y^2}\, dx\, dy. \quad f = f(r(x, y))$

$[\vec{n} = (-z_x, -z_y, 1), \quad z = z(x, y)]$

v. polje: $\iint_S \mathbf{V} \cdot d\mathbf{S} =$

$= \iint_S P dy dz + Q dz dx + R dx dy =$

$= \iint_S \mathbf{V} \cdot \vec{v}\, dS = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v]\, du\, dv$

$\vec{v} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}, \quad |\vec{r}_u \times \vec{r}_v| = \vec{n}$

Integralski izreki

Greenov, Stokesov, Gaussov :

$\oint_C (P\, dx + Q\, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\, dy$

C - rob pozitivno orientiranega $\bigcirc$ sklenjenega območja D

$\oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S rot\mathbf{V} \cdot d\mathbf{S} =$

$= \iint_D (rot\vec{V}, \vec{r}_u, \vec{r}_v) du\, dv$

C je rob površine S, in D je projekcija na ravnino

$\iint_S \vec{\mathbf{V}} \cdot d\mathbf{S} = \iiint_W div\vec{\mathbf{V}}\, dx\, dy\, dz$

S - zaprt površina, ki je rob telese W, usmerjena navzven

Gauss: $\iiint_S \vec{v} \cdot d\mathbf{S} = \iiint_V div\, \vec{v}\, dx\, dy\, dz;$

$vol(V) = \iiint_V dx\, dy\, dz = \frac{1}{3} \iiint_S (x, y, z) \cdot d\vec{\mathbf{S}}$

$\iiint_S \frac{\partial f}{\partial v} dS = \iiint_V \Delta f\, dx\, dy\, dz;$

$\iiint_S \vec{v} \cdot d\mathbf{S} = 0$ če je polje solenoidalno

Stokes: $\oint_{C=\partial S} \vec{v} \cdot d\vec{r} = \iint_S rot\, \vec{v} \cdot d\mathbf{S}$

$\iiint_S rot\, \vec{v} \cdot d\mathbf{S} = 0$ če je polje irrotacionalno

Kompleksna analiza

Analitičnost in odvedljivost in harmoničnost

Analitična funkcija: $f(z) = u(x, y) + iv(x, y)$

C-R pogoj: $u_x = v_y \quad u_y = -v_x \quad u - x = \frac{\partial u}{\partial x}$

$u = \int v_y\, dx = - \int v_x\, dy + C; \quad$ "je Analitična"

Pogoj Harmoničnosti:

$\nabla^2 u = \nabla^2 v = 0, \quad w_{xx} + w_{yy} = 0$

Elementarne komp. fun.

$z = x + iy; \quad |z| = \sqrt{x^2 + y^2};$

Polarni kot: $\arg z = \arctan\left(\frac{y}{x}\right)$

$e^z = e^x(\cos y + i \sin y)$

$\log z = \log |z| + i(\arg z + 2\pi k)$

$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = \sin x \cosh y + i \cos x \sinh y$

$\cos z = \frac{e^{iz} + e^{-iz}}{2} = \cos x \cosh y - i \sin x \sinh y$

$\sinh z = \frac{e^z - e^{-z}}{2}, \cosh z = \frac{e^z + e^{-z}}{2}$

lastnosti komp. funkcij

$e^{i\phi} = \cos \phi + i \sin \phi, \quad z = x + iy$

$|e^{iy}| = 1; \quad z = |z|e^{i\phi}, \quad z^\alpha = e^{\alpha \ln z}$

$\sin(\phi) = \frac{e^{i\phi} - e^{-i\phi}}{2i}; \quad \cos(\phi) = \frac{e^{i\phi} + e^{-i\phi}}{2}$

$z = e^{i\phi}; \quad d\phi = dz/iz; \quad \int_0^{2\pi} f(\sin(\phi), \cos(\phi)) d\phi$

$\frac{z - z^{-1}}{2i} = \sin(\phi); \quad \frac{z + z^{-1}}{2} = \cos(\phi)$

Integral Kompleksne funkcije

$\int_C f(z)\, dz = \int_C (u + iv)(dx + i dy) =$

$= \int_C (u\, dx - v\, dy) + i \int_C (v\, dx + u\, dy)$

$= \int_{t_a}^{t_b} f(z(t)) \cdot \dot{z}(t)\, dt;$ prevod v simpl. INT

Cauchjeva form.: $f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$

!Integral analitične. fun. ni odvisen od poti!

Integrali
Pravila integriranja

Substitucija: $\int f(g(x))g'(x)dx =$

$= \int f(u)du, \quad u = g(x)$

Po delih: $\int u\,dv = uv - \int v\,du$

Parcialni ulomki: $\frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x - a_i)^{n_i}}$

Elementarni integrali

$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$

$\int \frac{1}{x} dx = \ln|x| + C$

$\int e^x dx = e^x + C$

$\int \sin x dx = -\cos x + C$

$\int \cos x dx = \sin x + C$

$\int \frac{1}{1+x^2} dx = \arctan x + C$

$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$

$\int \sqrt{x} dx = \frac{2}{3}x^{3/2} + C$

Odvodi
Pravila odvajanja

$$\begin{aligned}(f+g)' &= f' + g' \\ (fg)' &= f'g + fg' \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} \\ (f(g(x)))' &= f'(g(x)) \cdot g'(x)\end{aligned}$$

Elementarni odvodi

$\frac{d}{dx}x^n = nx^{n-1}$

$\frac{d}{dx}e^x = e^x$

$\frac{d}{dx}\ln x = \frac{1}{x}$

$\frac{d}{dx}\sin x = \cos x$

$\frac{d}{dx}\cos x = -\sin x$

$\frac{d}{dx}\tan x = \sec^2 x$

$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$

$\frac{d}{dx}\arcsin x = \frac{1}{\sqrt{1-x^2}}$

Trigonometrske relacije

$\sin^2 x + \cos^2 x = 1$

$\tan x = \frac{\sin x}{\cos x}$

$\sin 2x = 2 \sin x \cos x$

$\cos 2x = \cos^2 x - \sin^2 x$

$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$

$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$

$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$

Adicijski izrek za sin,cos :

$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$

$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$